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SQL Exerice 2: Aggregate functions and grouping

1. COUNT( ) function

SQL query to find the total number of employees in the company

```
SELECT  COUNT(*)  AS  total_employees
FROM  employees_db
```

outcome :

total_employees
10

2. SUM( ) function

SQL query to find the total salary paid to all employees in IT

```
SELECT  SUM(salary)  AS  total_IT_salary
FROM  employees_db
WHERE  department = 'IT'
```

total_IT_salary
220 000

3. AVG( ) function

SQL query to calculate the average salary of employees in HR departm

```
SELECT  AVG(salary) AS  average_HR_salary
FROM  employees_db
WHERE  department = 'HR'
```

salary - min - salary
49 500

4. <u>MIN() and MAX() functions</u>

SQL query to find highest and lowest salary in the company

```
SELECT MIN(salary) AS lowest_salary,
       MAX(salary) AS highest_salary.
FROM employees
```

lowest_salary	highest_salary
48 000	62 000

5. <u>GROUP BY Statement</u>

SQL query to find group employees by department and display total salary paid to each department.

```
SELECT department,
       SUM(salary) AS total_salary
FROM employees
GROUP BY department
```

department	total_salary
IT	220 000
HR	99 000
Finance	119 000
Marketing	105 000

6. GROUP BY and COUNT()

SQL query to count how many employees work in each city

```
SELECT    city,
          COUNT(*) AS total_employees
FROM employees
GROUP BY  city
```

City	total_employees
New York	2
Chicago	3
Los Angeles	2
San Francisco	2
Houston	1

7. GROUP BY and ORDER BY

SQL query to group employees by department, calculate the average salary in each department and order results in descending order of average salary.

```
SELECT    department
          AVG(salary) AS avg_salary
FROM employees
GROUP BY  department
ORDER BY  AVG(salary)  DESC
```

department	avg_salary
Finance	59 500
IT	55 000
Marketing	52 500
HR	49 500

8. HAVING Clause

SQL query to find departments where the total salary paid exceeds 100 000 (use GROUP BY and HAVING)

```
SELECT    department
          SUM(salary) AS total_salary
FROM      employees
GROUP BY  department
HAVING    SUM(salary) > 100 000
```

Department	total_salary
IT	220 000
Finance	119 000
Marketing	105 000

9. Combining GROUP BY, HAVING and ORDER BY

SQL query to list cities where more than one employees works, ordered by the number of employees in descending order

```
SELECT    city
          COUNT(*) AS total_employees
FROM      employees
GROUP BY  city
HAVING    COUNT(*) > 1
ORDER BY  COUNT(*)    DESC
```

City	total_employees
Chicago	3
New York	2
Los Angels	2
San Francisco	2

10 Combining Aggregate functions

SQL query to find the department with highest average sal

```
SELECT department
       AVG(salary) AS average-salary
FROM employees
ORDER BY AVG(salary) DESC
LIMIT 1
```

department	average-salary
Finance	59 500